Ayush Kumar

ML Engineer | AI Researcher

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SUMMARY

Independent AI researcher from India working on multi-agent LLM systems, synthetic data generation for RLHF/DPO, and ML engineering, including INT8 quantization and MLOps. At the AIRI Institute of Artificial Intelligence (TSU x AIRI), developed Auto-ChemInstruct, an autonomous four-agent pipeline for generating physically verified chemical datasets without human annotators. Manuscript in preparation for NeurIPS Datasets & Benchmarks 2026. Placed in the Top 30 of the Wunder Fund RNN Challenge global ranking by implementing custom INT8 quantization in PyTorch and reducing memory usage by 75% for a 10-model ensemble. Seeking research or ML engineering roles in Generative AI, agentic systems, or scientific computing.

EDUCATION

National Research Tomsk State University

Tomsk, Russia | 2023 - 2027

Degree program: Mathematics and Computer Science

Specialization: Computational Mathematics and Computer Modeling

Faculty: Faculty of Mechanics and Mathematics

Department: VMiKM (Computational Mathematics and Computer Modeling)

Qualification: Computational Mathematician

Selected coursework and focus

- Machine Learning: applied ML and artificial neural networks (Python, C++), ML fundamentals, and deep learning.
- Applied Science and Natural Sciences: computational analysis of natural science problems, computational mechanics, and numerical methods in physics and biology.
- Computational Mathematics: advanced numerical methods, computer modeling, and numerical solutions for industrial mathematics.
- High-Performance Computing: parallel programming and parallel computing methods.
- Mathematical Modeling: differential equations and finite-difference schemes for ecology, medicine, and mechanics.

EXPERIENCE

Big Mathematical Workshop (BMM), Novosibirsk State University

Research Team Lead

July 2026 - Present | Novosibirsk, Russia

- Working on AlignSSL-SV, a neural network system for detecting genomic mutations, especially deletions, from sequencing read alignments. Deletions are losses of DNA segments in the genome.
- Compared an image-free alignment-tensor representation with a DeepSV-style RGB-pileup baseline; the project uses multi-scale CNNs, a transformer encoder, and a calibrated deletion classifier.
- Ran experiments with masked alignment modeling, varying amounts of labeled data, cross-population evaluation, and hard-negative controls on 1000 Genomes data.
- Led the research team and maintained a reproducible Python pipeline for data extraction, self-supervised pretraining, fine-tuning, model evaluation, result analysis, and preprint preparation.
- The study is ongoing; a preprint is in preparation.

AIRI Institute of Artificial Intelligence (TSU x AIRI)

AI Research Intern

March 2026 - May 2026

- Investigated the data bottleneck for chemical LLMs. Existing datasets, including ChemCoTBench, have positive-label bias, require costly manual annotation, and lack explanations for failed reactions.
- Developed Auto-ChemInstruct, an autonomous four-stage multi-agent pipeline for generation, verification, reflection, and compilation. The pipeline creates large, physically verified datasets for RLHF/DPO without human annotators.
- Reduced positive-label bias by using RDKit and xTB to turn failed reactions into causal chain-of-thought failure traces and negative training examples.
- Orchestrated the pipeline through Maestro with a research-team topology of a GI agent, analyst, and literature reviewer. Used GigaChain for multimodal RAG with PubChem and Materials Project, plus GigaEvo for MAP-Elites prompt optimization through LangGraph.
- Co-authored a paper on scalable agent-based synthetic data generation for chemistry. Manuscript in preparation for NeurIPS Datasets & Benchmarks.

Wunder Fund RNN

Independent Machine Learning Researcher

September 2025 - January 2026

- Designed and deployed an inference pipeline for high-frequency trading market-state forecasting. Placed in the Top 30 of the Wunder Fund Challenge global ranking.
- Developed a custom dynamic INT8 quantization pipeline in PyTorch for deployment on a single CPU core with a memory limit under 20 MB.
- Reduced memory usage by 75%, allowing a 10-model ensemble to run within the resource budget.
- Designed a Hive Mind ensemble of deep residual GRUs and LSTMs with Mish activations and Squeeze-and-Excitation blocks.
- Set up MLOps practices including automated artifact serialization, custom Huber and Pinball losses, and cross-validation to control overfitting on small financial datasets.

Turing AI

Mathematics Problem Creator and Programmer

April 2025 - August 2025

- Created original, challenging mathematics and algorithmic problems similar to Project Euler for training AI models in reasoning and logic.
- Wrote reference solutions in C++ and Python under strict time and memory limits, including solutions with O(n log n) complexity.
- Designed test cases and checked edge conditions to verify solution correctness and stability.
- Wrote step-by-step explanations and hints for each solution.

PROJECTS

AlignSSL-SV - Discovering Neural Networks for Detecting Mutations in Genomic Data GitHub | Big Mathematical Workshop 2026, Great Math Track

- AlignSSL-SV detects deletions, or losses of DNA segments, in sequencing data using neural networks. The model converts read alignments into tensors and predicts whether a deletion is present in a genomic region.
- The stack and evaluation include multi-scale CNNs, a transformer encoder, masked alignment modeling, temperature scaling, and MC dropout; experiments use 1000 Genomes data with cross-population evaluation, label-efficiency sweeps, and hard-negative controls.
- Research code is under active development; preprint in preparation.

Auto-ChemInstruct - Physics-Verified Chemical Dataset Generator GitHub | Manuscript in preparation

- Designed a four-agent autonomous pipeline: a hypothesis agent generates reactions, a verification agent uses RDKit and xTB as physical oracles, a reflection agent builds causal chain-of-thought traces, and a compilation agent creates preference pairs for DPO/RLHF.
- Failed reactions are retained. The reflection agent explains causes such as steric hindrance and high energy barriers, producing negative examples that standard datasets often lack.
- Implemented GigaChain for literature processing, embeddings for SMILES and IUPAC, and calls to PubChem and Materials Project.
- Integrated the MAP-Elites evolutionary loop from GigaEvo to optimize agent prompts through LangGraph mutations.
- Coordinated the workflow through Maestro with a virtual research team of a GI agent, analyst, and literature reviewer, keeping dataset generation in organic chemistry, cheminformatics, and materials science reproducible and auditable.

AuthCore - Custom Authentication and RBAC System GitHub

- Developed a stateless Python authentication system using JSON Web Tokens (JWT) for session management.
- Implemented password hashing with bcrypt.
- Designed a role-based access control (RBAC) model to protect REST API endpoints.
- Created middleware to check user permissions against a PostgreSQL database.
- Automated deployment and maintenance through Shell scripts.

Efficient Neural HFT (Market State Forecasting) GitHub | Hugging Face demo

- Designed a Hive Mind ensemble of deep residual GRUs and LSTMs with Mish activations and Squeeze-and-Excitation blocks for financial time-series forecasting.
- Developed a custom dynamic INT8 quantization pipeline in PyTorch for single-core deployment with a memory limit under 20 MB.
- Reduced memory usage by 75%, allowing a 10-model ensemble to run within a constrained resource budget.
- Implemented automated artifact serialization and custom Huber and Pinball financial losses.

CERTIFICATIONS

- Deep Learning Specialization - DeepLearning.AI (January 2024)
- Neural Networks and Deep Learning - DeepLearning.AI (December 2023)
- IBM Machine Learning - Coursera (July 2023)
- Python Pro Bootcamp (100 Days of Code) - Udemy (December 2022)
- Full-Stack Web Development Bootcamp - Udemy (May 2022)

SKILLS

- ML and Deep Learning: PyTorch, Transformers, sequence models (RNN, LSTM, GRU), LLMs, quantization, pruning, model optimization
- Generative AI and Agentic Systems: multi-agent orchestration, LangChain, LangGraph, RAG, prompt engineering, synthetic data generation, RLHF/DPO, fine-tuning
- AI Engineering and MLOps: LLM integration, workflow design, agentic AI pipelines, deployment, monitoring, evaluation, CI/CD for AI
- Data Science and Scientific Computing: Pandas, NumPy, SciPy, Matplotlib, SQL
- Programming: Python (Expert), C++ (Intermediate), SQL, Bash
- Infrastructure and Tools: Git, Docker, Linux, FastAPI
- AI Development Tools: Claude Code, Cursor, OpenClaw, Hermes Agent, GitHub Copilot, Codex, OpenCode, Cline, n8n, CLI agents
- Languages: English (Native/Fluent), Russian (Professional)

AWARDS

- Big Mathematical Workshop 2026, Great Math Track (Novosibirsk State University): certificate of participation for work on "Discovering Neural Networks for Detecting Mutations in Genomic Data"; served as Research Team Lead, July 9-21, 2026
- Wunder Fund RNN Challenge - Market State Forecast: 28th place in the global ranking, 2025
- T-Bank Code Gym 2025 (Tomsk): 3rd place, team 40 + 40 + 1 = 1001
- Code Sport Tomsk - T-Education 2026: 4th place, Vibe Compilers team
- AvitoTech ML CUP 2025 - Open Data Science (ods.ai): certificate of successful completion